Subqueries

Subqueries

- Asubqueryisa“**querywithina query”.**
- A SELECT statement nested within another SELECT statement.

```
SELECT …                    ←——— Main query
FROM …
WHERE ….   ( SELECT …
            FROM ….
           WHERE….. );      ←——— Subquery
```

TypesofSubqueries

1. Single–row subqueries: Return only one row of result. They mark the usage of single row comparison operators, when used in WHERE conditions.
2. Multirow subqueries: Return more than one row of result. Theymake use of multiple row comparison operators like IN, ANY, ALL. There can be sub queries returning multiple columns also.
3. Multiple-columnsubqueries:returnsmorethanonecolumn.
4. Correlated Subqueries: Correlated subqueries depend on data provided by the outer query. This type of subquery also includes subqueries that use the EXISTS operator to test the existence of data rows satisfying specified criteria.

Single RowSubquery

- Listallemployees whosesalaryisgreaterthanthatofSMITH. Sol:
- FindthesalaryofSMITH.

  Selectsalfromempwhereename=‘SMITH’;
- Usethissalarytofindalltheemployees.

  selectempno,enamefromempwheresal>(selectsalfrom empwhereename=‘SMITH’);

Example-2

- Listallemployeesworkinginsamedepartmentinwhich “SMITH” is working.

Sol:FindthedepartmentnumberofSMITH.

selectdeptnofromempwhere ename=‘SMITH’

- Usethisdepartmentnumbertofindwhichemployeesare working.

select empno, ename from emp where deptno = (select deptno from emp where ename=‘SMITH’);

Example-3

- Listallemployees workingin“ACCOUNTING” department.

Sol:

- Findthedepartmentnumberof department“ACCOUNTING”

  Selectdeptnofromdeptwhere dname=‘ACCOUNTING’;

- Use this department number to list all the employees workingin same department.

  Select empno, ename from empwheredeptno = (Select deptno from dept where dname=‘ACCOUNTING’);

Contd..

1. Listtheemployeesworkingunder“KING”.
2. Find the employees whose department is same as “ALLEN”and salary is less than the salary of “BLAKE”.
3. Tofindtheemployeewithhighestsalary.
4. Tofindtheemployeeswhoearnmorethantheaveragesalary oftheir company.
5. Tofindthedepartmentswhoseaveragesalaryismorethan average salary of department 10.
6. Findthedepartmentwithlowestaveragesalary.

Contd..

7. FindthedetailsofemployeeworkinginDALLAS.
8. Findwhoisgetting theminimumsalaryindepartment10.
9. FindtheemployeeswhoareseniortoSMITH.
10. Findoutwhoaregrade5 Employees.
11. Whoaretheemployeesworkingunder BLACK.
12. Findthemanagerbeingreportedbymaximumnumberof employee.
13. Find the department number having highest number of employees.

Contd..

- Whatwillbetheoutput??

- SELECTename,sal,deptnoFROMemp
  WHEREdeptno=(SELECTdeptnoFROMdeptWHERE dname IN (‘SALES’,’ACCOUNTING’));

- ERROR:single-rowsubqueryreturnsmorethanonerow

Multi-rowSubqueries

- Returnmorethanonerow
- Uses a multiple-row comparison operators suchasIN, ANY,ALL , EXISTS.
- Most commonlyusedoperatorisIN

Example

- ListalltheemployeesworkinginSALESandACCOUNTING departments.

- SELECTename,sal,deptnoFROMemp

  WHEREdeptnoIN(SELECTdeptno FROMdeptWHERE dnameIN(‘SALES’,’ACCOUNTING’));

Use ofANYoperator

- Listtheemployeeswhoarenotindepartment10andwhose salary is less than that of any employee of department 10.

- SELECTename,sal,deptnoFROMemp

  WHEREsal<ANY (SELECTsalFROMempWHERE deptno=10)ANDdeptno<>10;

UseofALLoperator

- List the employees who are not in department 10 and whose salaryislessthanthesalaryofalltheemployees ofdepartment 10.
- SELECTename,sal,deptnoFROMemp
- WHEREsal<ALL(SELECTsalFROMempWHERE deptno=10)ANDdeptno<>10;

MultipleColumnSubQueries

- Amultiple-column subquery returns more than one column to the outer query.
- Syntax:

```
SELECT col1, col2, ...
FROM table
WHERE (col1, col2, ...) IN
                ( SELECT (col1,col2, ...)
                FROM table
                WHERE condition );
```

Example:

- Findsthehighest-paidemployee ineachdepartment.
  SELECT deptno, ename, sal FROM emp
  WHERE(deptno,sal)IN(SELECTdeptno,max(sal)
  FROMempGROUPBYdeptno);

- Findsthelowest-paidemployeeofeachmanager.
  SELECTempno,ename, salFROMemp
  WHERE(mgr,sal)IN(SELECTmgr,min(sal)FROMemp
  GROUP BY mgr);

CorrelatedSubquery

- Acorrelated subqueryis onewherethe innerquerydepends on values provided by the outer query. This means that in a correlated subquery, the inner query is executed repeatedly, once for each row that might be selected by the outer query.
- Correlatedsubqueriescanproduceresulttablesthatanswer complexmanagement questions.

Contd..

- Querytolistemployeename,department,andthehighestsalary in that department.
- SELECTename,deptno,

  (**SELECTmax(sal)FROMempe1**

  WHEREe1.deptno=e.deptno)HSALFROMempe;

Example

- List all employees whose salaries exceed their department average.

  SELECTdeptno,ename,salFROMempx

  WHEREsal>(SELECTAVG(sal)FROMemp

  WHEREdeptno=x.deptno);

- For each row of the emp table, the parent query uses the correlated subquery to compute the average salary for members of the same department.

UsingExistsOperator

- Used with correlated subqueries totest forexistenceofrows in the result set of the subquery.
- ReturnsTRUE,ifthesubqueryreturnsatleastonerow.
- ReturnsFALSE,ifasubqueryrowvalueisnotfound.

Example

- Findallemployeeswhohave atleastone personreporting to them.
- SELECTempno,enameFROMempe1WHEREEXISTS(SELECT‘X’FROMempWHEREmgrid=e1.empno);
- Theinnerquerydoesn’tneed toreturnanyvalue.
- Thesearchininnerquery doesn’tcontinuewhenatleastone match is found.

AlternativeExistsOperator

- SELECTempno,enameFROMemp
- WHEREempno**IN(SELECTmgr**FROMempWHERE mgr IS NOT NULL);

Example

- Selectdeptno,dnamefromdept
- wheredeptnoIN(selectdeptnofromemp);

- WritetheabovequeryusingEXISTS.

- Selectdeptno,dnamefromdeptd
- whereEXISTS(select‘X’fromempewheree.deptno=d.deptno);

DMLstatement

- DELETEtheemployeeswhosesalaryisbelowtheaverage salary in their department .

- DELETE FROMempe
- WHEREsal<(SELECTAVG(sal)FROMemp
- WHEREdeptno=e.deptno);

UpdateStatement

- UPDATEthesalaryofallemployeestothemaximumsalaryin the corresponding department

- UPDATEemp e1 SETsal =(SELECTMAX(sal) FROMemp e2 WHERE e1.deptno=e2.deptno);

Example

- Listallemployees whodonothaveanysubordinates.

SELECT ename FROM emp

WHEREempnoNOTIN(SELECTmgrFROM emp);

Or

SELECTenameFROM emp

WHEREempnoNOTIN(SELECTmgrFROMempWHERE mgr **IS NOT NULL**